Slabs:
pieces at girth five without the lemma on rigid sets

A fifth research note on “The maximum number of cycles in a graph of large rank”

Fable 5.2, for KNT — Research note of 13 November 2026

On 12 November 2026 I wrote a formal proof of [P, Theorem 7.1] at girth five in Lean. Here [P] is the paper of 36 pages (commit `805e2ba`). The formal proof of [P, Lemma 7.6] (pieces at girth five) has an inner step that is not in the paper. KNT asked for a draft of this step as text, and for a measurement. This note has the draft in two forms, with complete proofs, and the numbers. The manuscript is not changed.

The result in five sentences

1. In the proof of [P, Lemma 7.6] a part M of the piece is long and thin, and its boundary vertices are at its two ends. So M has many disjoint slabs across a shortest path, and for one slab B every vertex of a given set U is on one of the two sides of B.
2. For this slab a U-join of M meets $M[B]$ in an even subgraph or in a join of two vertices. So a rigid set U gives a full pair in $M[B]$, and [P, Lemma 7.4] applies. This replaces [P, Lemma 7.5] (rigid sets) with its tree, its segments and its count.
3. Form *slabs*: [P, Lemma 7.5] is removed and [P, Lemma 7.4] stays. Measured in a copy: 1288 typeset lines in the place of 1305, and 36 statements in the place of 37; 36 pages and 15 figures as before.
4. Form *slabs and rails*: in addition [P, Lemma 7.4] gets a proof without ladders, by a count beside a shortest path and a fact on seven integers. Section 7 then uses no result of Section 5 of the paper, and the text of Section 7.2 is the formal proof. Measured: 1308 lines, so three lines more than now.
5. The formal proof in Lean has the second form, with other constants. The first form differs from it in one proof, that of [P, Lemma 7.4]. I recommend the form *slabs* if the aim is fewer steps, and the form *slabs and rails* if the aim is a text that the formal proof follows. The gain is small in both forms.

1 The measurement

The script `research/apply_slabs.py` changes a copy of the folder `exact-maximum`; with the argument `rails` it applies the second form. The copies are `research/trial-slabs.pdf` and `research/trial-slabs-rails.pdf`. Both build without an overfull box and without an undefined reference. A typeset line is a line of `pdftotext -layout` that is not empty and is not a running head.

	pages	lines	Section 7	Section 7.2	statements	figures
the paper now	36	1305	378	121	37	15
form *slabs*	36	1288	361	103	36	15
form *slabs and rails*	36	1308	377	120	36	15

In both forms the figure of the tree through a rigid set is replaced by a figure of the slabs (Figure 1). The statements after [P, Lemma 7.4] get a number that is smaller by one. No other section changes, with the exception of one sentence on the formal proof in [P, Section 1.4].

What the two forms change in the proof, step by step:

the paper now	form *slabs*	form *slabs and rails*
[P, Lemma 7.4]: a full pair at girth five is small; by the reduction [P, Lemma 5.2] and the ladders of [P, Lemma 5.6]	the same	a rigid pair at girth five is small; by a shortest path, a forest beside it, and seven integers
[P, Lemma 7.5]: a thin multigraph with a rigid set is small; a smallest tree through the set, its segments, a far point, an exchange step, two cases, a count	no such lemma	no such lemma
[P, Lemma 7.6]: groups of layers, then [P, Lemma 7.5] and the product	groups of layers, slabs, side components, the same two cases, the product	the same as in the form *slabs*

2 The new proof of Lemma 7.6

NOTATION | what the proof uses from the paper

- N is a connected subcubic multigraph. Its vertices of degree less than three are its boundary vertices, its deficit $d(N)$ is the sum of $3 - \deg v$ over all vertices, and $|V(N)| = 2\beta(N) - 2 + d(N)$; this is [P, equation (7.1)].
- For an even set T of boundary vertices, a T-join is a set of edges in which the vertices of T have odd degree and all other vertices have even degree. The forest proportion of T is the proportion of the T-joins that are forests, and T is rigid if every T-join is a forest.
- [P, Lemma 7.3] (product). Let pairwise disjoint connected sets of vertices of degree three, without a vertex of T, induce multigraphs in which every even set of boundary vertices has forest proportion at most f_i. Then the forest proportion of T in N is at most the product of the f_i.
- [P, Lemma 7.4] (full pairs at girth five). Let M be a connected subcubic multigraph of girth at least five, and let $x \neq y$ be vertices of degree at most two such that every $\{x, y\}$-join of M is a path. Then $|V(M)| < 15\, d(M)$.
- B is the constant of [P, Section 7.1]: in a piece with k boundary vertices and a forest proportion of at least $\frac{3}{5}$, every distance layer from a boundary vertex has fewer than $B(k+1)^2$ vertices. This is the step "A marker" in the proof of [P, Lemma 7.2].

Lemma 2.1 (Pieces at girth five; this is [P, Lemma 7.6]). For every integer $k \geq 1$ there is a number $F_0(k)$ with the following property. Let N be a connected subcubic multigraph of girth at least five with at most k boundary vertices and at least $F_0(k)$ vertices. Then every set of boundary vertices of even size has forest proportion less than $\frac{3}{5}$.

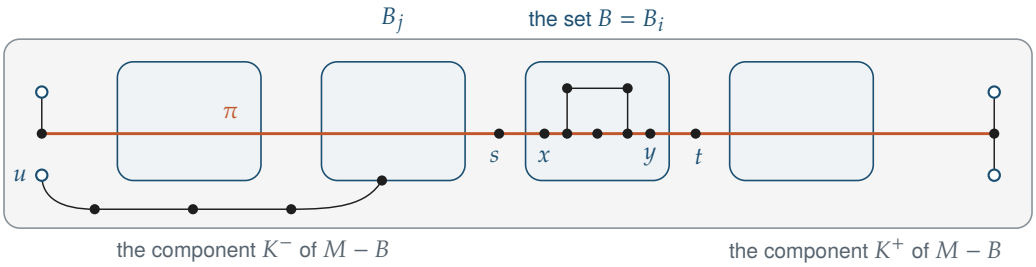


Figure 1. The proof of Lemma 7.6. The part M with the path π (thick), four of its slabs, and the vertices of U (circles) in the first and in the last layer. The vertex u is in a side component of $M - B_j$; for the slab B it is in K^-. If $K^- \neq K^+$ and each of them contains an odd number of vertices of U, then a U-join of M meets $M[B]$ in an $\{x, y\}$-join.

Proof. Put $W = B(k+1)^2$, $\tau = 60\,W$, $h = 2W\tau + 2$ and $F_0(k) = Wh\,(k + 2^{Wh} + 2)$. Suppose that some even set T of boundary vertices has forest proportion at least $\frac{3}{5}$. If T is empty, then N has no cycle and fewer than $2k$ vertices. In the other case fix a boundary vertex o, and define the layers as in the proof of Lemma 7.2. The step "A marker" of that proof uses no girth, and it shows that each layer has fewer than W vertices. So N has at least $h\,(k + 2^{Wh} + 2)$ layers. Divide the layers $1, 2, \ldots$ into groups of h consecutive layers. At least 2^{Wh} groups contain no boundary vertex. In such a group let M be a component of the subgraph induced by the group that has a vertex v in the last layer of the group. A shortest path from o to v has one vertex in each layer of the group; let π_j be its vertex in the jth of these layers. The path $\pi = \pi_1 \cdots \pi_h$ lies in M. The vertices of M have degree three in N, and an edge that leaves M starts in the first or in the last layer of the group. So the boundary vertices of M are in these two layers, there are fewer than $2W$ of them, and the rank of M is less than Wh. We show that no even set of boundary vertices of M is rigid (Figure 1).

Slabs. For $1 \leq i \leq 2W$, the ith *slab* consists of the layers $2+(i-1)\tau, \ldots, 1+i\tau$ of the group. Let B_i be the vertex set of the component of the subgraph of M induced by this slab that contains the τ vertices of π in the slab. Let $s_i x_i$ be the edge of π that enters B_i and $y_i t_i$ the edge that leaves it, with $x_i, y_i \in B_i$. The vertices of B_i have degree three in M, and an edge of M that leaves B_i starts in the first or in the last layer of the slab. So $d(M[B_i]) < 4W \leq |B_i|$. Hence $M[B_i]$ has a cycle, by (7.1), and $x_i \neq y_i$ are boundary vertices of $M[B_i]$. Call a component of $M - B_i$ a *side component* if it contains neither s_i nor t_i. The vertices of π outside B_i are joined to s_i or to t_i in $\pi - B_i$. A set B_j with $j \neq i$ is connected, avoids B_i and contains a vertex of π. So a side component of $M - B_i$ meets no set B_j.

Side components. Let K be a side component of $M - B_i$, and let $j \neq i$. Some vertex of K has a neighbour in B_i, because M is connected. So $K \cup B_i$ is connected, avoids B_j and contains a

vertex of π. Hence it lies in the component of s_j or of t_j in $M - B_j$. So a vertex of M is in a side component of $M - B_i$ for at most one i.

No rigid set. Let U be an even set of boundary vertices of M. It has fewer than $2W$ vertices, and none of them is in a slab. So there is an i such that no vertex of U is in a side component of $M - B_i$. Write B, s, x, y, t for B_i, s_i, x_i, y_i, t_i, and let K^- and K^+ be the components of $M - B$ that contain s and t. Every vertex of U is in one of them. Put $U^- = U \cap V(K^-)$. Suppose first that $K^- = K^+$ or that $|U^-|$ is even. Then $M - B$ contains a U-join, and its union with a cycle of $M[B]$ is a U-join that is not a forest. Suppose now that $K^- \neq K^+$ and that $|U^-|$ is odd, and let J be an $\{x, y\}$-join of $M[B]$. Add the edges sx and ty, a join of K^- for the set U^- with s added or removed, and a join of K^+ for the set $U \setminus U^-$ with t added or removed. The result is a U-join of M. If U is rigid, then J is a forest, and hence a path. Then Lemma 7.4 gives $|B| < 15\, d(M[B]) < 60\, W = \tau$, and this is false.

So each even set of boundary vertices of M has forest proportion at most $1 - 2^{-Wh}$. By Lemma 7.3, the forest proportion of T is at most $(1-2^{-Wh})^{2^{Wh}} < \frac{1}{2}$. This is a contradiction. ■

The proof begins and ends as the proof in the paper does. The steps "Slabs", "Side components" and "No rigid set" are in the place of the reference to [P, Lemma 7.5]. The paper has $h = 10^4 W^3$; here $h = 120\, W^2 + 2$.

3 A proof of Lemma 7.4 without ladders

An $\{x, y\}$-join that is a forest is a path. So the next lemma is [P, Lemma 7.4] with the constant 133 in the place of 15. With it, the proof of Lemma 2.1 takes $\tau = 532\, W$, and its last case ends with $|B| < 133\, d(M[B]) < 532\, W = \tau$.

Lemma 3.1 (Rigid pairs at girth five). Let M be a connected subcubic multigraph of girth at least five. Let $x \neq y$ be vertices of degree at most two such that every $\{x, y\}$-join of M is a forest. Then $|V(M)| < 133\, d(M)$.

Proof. Put $d = d(M)$. Let $A = a_0 a_1 \cdots a_\ell$ be a shortest path from x to y, and put $F = M - V(A)$. The edge set of A is an $\{x, y\}$-join, and so is its union with a cycle of F. So F is a forest. A vertex of F has at most one neighbour on A: two neighbours a_i and a_j with $i < j$ give $j - i \leq 2$, because A is a shortest path, and then a cycle with at most four edges. For the same reasons an inner vertex of A of degree three has a neighbour in F.

Call a vertex of F *plain* if it has degree three and a neighbour a_p with $0 < p < \ell$. A plain vertex has exactly two neighbours in F. A vertex of F that is not plain is a boundary vertex, or it is adjacent to x or y, or it has three neighbours in F. Let n_0, n_1 and n_3 be the numbers of these vertices. Then $n_0 \leq d - 2$ and $n_1 \leq 2$, because x and y are boundary vertices. The forest F has at most as many edges as vertices, so a count of edge ends gives $3n_3 + 2(|F| - n_0 - n_3) \leq 2|F|$, and $n_3 \leq 2n_0$. So fewer than $3d$ vertices of F are not plain.

No path of seven plain vertices. Suppose that $v_0 v_1 \cdots v_6$ is a path of plain vertices in F, and let

> a_{p_k} be the neighbour of v_k on A. Let $0 \le k < l \le 6$. Then $p_k \ne p_l$, because a_{p_k} has three neighbours only. Also $|p_k - p_l| \le l - k + 2$, because A is a shortest path and $a_{p_k} v_k \cdots v_l a_{p_l}$ is a path. And $|p_k - p_{k+1}| \ge 2$, because M has no cycle with four edges. Put $s_k = p_{k+1} - p_k$ for $0 \le k \le 5$. Then $|s_k|$ is 2 or 3, and $0 < |s_k + s_{k+1}| \le 4$. So two consecutive steps of the same sign are both 2 or both -2, and of two consecutive steps of different signs one has absolute value 2 and the other has absolute value 3. No three consecutive steps have the same sign, because $|p_{k+3} - p_k| \le 5$.
>
> If consecutive steps always have different signs, then $s_0 = s_2 = s_4$ and $s_1 = s_3 = s_5$. Then $p_5 - p_0 = 3s_0 + 2s_1 = 0$ if $|s_0| = 2$, and $p_6 - p_1 = 2s_0 + 3s_1 = 0$ if $|s_0| = 3$. In the other case let $s_k = s_{k+1} = 2\varepsilon$ with $\varepsilon = \pm 1$. Where these steps exist, $s_{k-1} = s_{k+2} = -3\varepsilon$ and $s_{k-2} = s_{k+3} = 2\varepsilon$. Then $p_{k-1} = p_{k+4}$ and $p_{k-2} = p_{k+3}$, so in each of these two pairs one index is outside the range $0, \ldots, 6$. This leaves $k = 0$ and $k = 4$. Let $k = 0$; the case $k = 4$ is the same with the order reversed. The first four steps are $2\varepsilon, 2\varepsilon, -3\varepsilon, 2\varepsilon$. If $s_4 = -3\varepsilon$, then $p_5 = p_0$. If $s_4 = 2\varepsilon$, then $s_5 = -3\varepsilon$ and $p_6 = p_1$. So F has no path of seven plain vertices.
>
> **The count.** Let w be a vertex of F such that all vertices at distance at most three from w in F are plain. Then w, its two neighbours in F, their other neighbours and the next two vertices form a path of seven plain vertices, because F is a forest. So every vertex of F is within distance three in F of a vertex that is not plain, and $|F| < 22 \cdot 3d$. An inner vertex of A of degree three has a neighbour in F, and a vertex of F has at most one neighbour on A. So A has at most $|F| + d$ vertices, and $|V(M)| \le 2|F| + d < 133\,d$. ∎

The bound of seven is the best one for this argument: the integers $5, 2, 4, 1, 3, 0$ satisfy the three conditions on the positions. The script `checks/rail_check.py` tests all sequences of steps (exit code 0).

4 The formal proof

The files are in `Erdos1016/ExactMaximum/GirthFive/` (commit `b7f9fc9`); the README of that folder has the same proofs in words.

text of this note	Lean declaration
the fact on seven integers	`GirthFive.rail_bound`
a vertex beside a shortest path has one neighbour on it; no path of seven plain vertices	`GirthFive.attach_unique`, `GirthFive.no_rail`
Lemma 3.1	`GirthFive.two_terminal_card_le`, `GirthFive.two_terminal_region`, with the bound $400\,(\|D\| + 2)$ for the set D of boundary vertices
the steps "Side components" and "No rigid set"	`GirthFive.side_of_other_slab`, `GirthFive.exists_word_two_terminals`, `GirthFive.region_not_rigid`
the step "Slabs"	`GirthFive.window_not_rigid`
Lemma 2.1	`GirthFive.exists_affine_bound_girth_five`

Two differences. The formal step "No rigid set" takes a slab without a side component that has an odd number of vertices of U; the text takes a slab without any vertex of U in a side

component, which the same count gives. The form *slabs* uses [P, Lemma 7.4] with its proof by ladders, and this proof has no formal counterpart; the formal proof has Lemma 3.1 in its place.

5 What I examined and did not use

- A slab that consists of all vertices of M in its layers. Then the graph outside the slab can have many components on each side, and a U-join meets the slab in a join of many vertices. The component of the path π and the count of side components give two vertices.
- A proof without a lemma on two vertices. A ladder has girth four, and the two ends of one rail form a rigid set. So some statement on rigid pairs at girth five is necessary, and a statement of this kind needs a number such as seven: six positions exist.
- A shorter arithmetic argument for the seven integers, by residues or by a count of the positions in a window. I found none. The case distinction has three cases.
- The lemma on rigid pairs inside the slab, where the path π is given and the layers bound the positions. The proof does not become shorter, because the condition on a shortest path is the one that the argument uses.